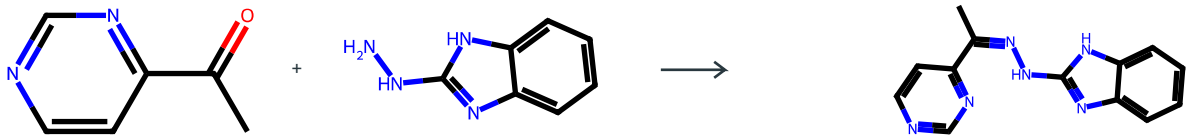
Performed experimental precedent for generated complete reactions

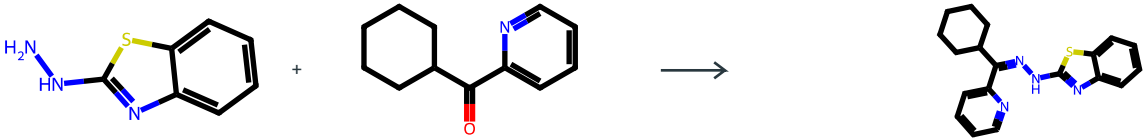
Selected cases; complete net chemistry, not experimentally established electron-arrow sequences.

1. Tier A · US7112680B2:example5:pyrimidinyl-benzimidazole-hydrazone



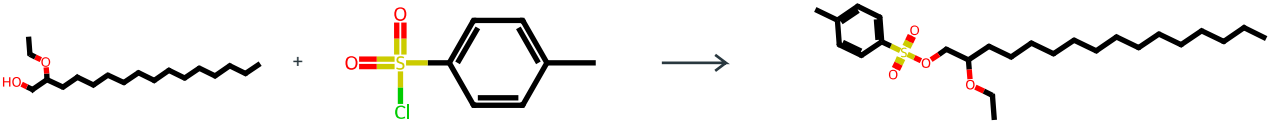
4-Acetylpyrimidine 0.412 g and 2-hydrazinobenzimidazole 0.50 g, each 3.37 mmol, in methanol 15 mL with five drops acetic acid; reflux to TLC completion, cool overnight near 5 C, filter/recrystallize. 0.65 g, 76%; elemental analysis/NMR reported. No E/Z assignment imposed.

2. Tier A · US7112680B2:example4:cyclohexyl-benzothiazole-hydrazone



Cyclohexyl(2-pyridyl)ketone 0.77 g/4.05 mmol and 2-hydrazinobenzothiazole 0.60 g/4.02 mmol, methanol 15 mL, five drops acetic acid; reflux 15 h, cool overnight near 5 C, filter and ethanol/water recrystallization. 1.09 g, 80%, elemental analysis/NMR; geometry not specified.

3. Tier A · US4046914:example4:ethoxyhexadecyl-tosylate



2-Ethoxyhexadecan-1-ol 1.72 g and tosyl chloride 1.5 g in pyridine 2.3 mL, room temperature overnight. 2.47 g crude product with only minor TLC impurities; no isolated percentage yield inferred from crude mass or later glycerol products.

Source/training overlap unknown; an experimentally documented transformation is not a tested whole route.